

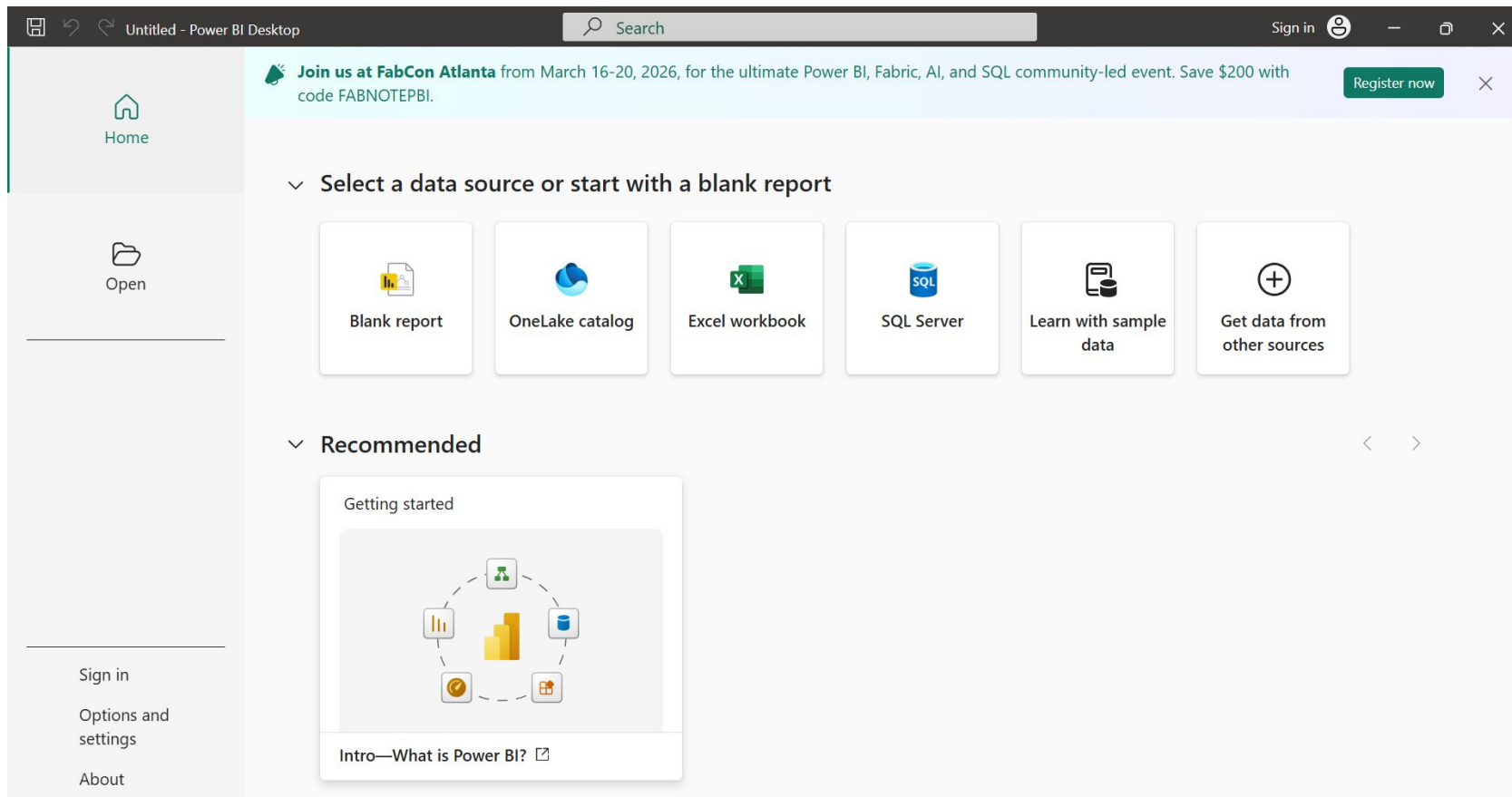
Introduction to Business Intelligence & Power BI

Justinian L. Melecio

Instructor: Joseph A. Vistal

IS 107 - Business Intelligence

PART 1 | Step 1: Open Power BI Desktop



PART 1 | Step 2: Load the Dataset

BI_LW1_Basic_Sales_Data.csv

File Origin: 1252: Western European (Windows) | Delimiter: Comma | Data Type Detection: Based on first 200 rows

Date	Product	Category	Region	Sales
1/1/2024	Product A	Electronics	East	175
1/2/2024	Product B	Furniture	South	378
1/3/2024	Product A	Electronics	North	842
1/4/2024	Product E	Furniture	North	582
1/5/2024	Product A	Electronics	South	245
1/6/2024	Product B	Furniture	North	667
1/7/2024	Product E	Furniture	West	353
1/8/2024	Product B	Furniture	East	609
1/9/2024	Product A	Electronics	West	313
1/10/2024	Product C	Office Supplies	South	434
1/11/2024	Product B	Furniture	North	494
1/12/2024	Product A	Electronics	North	539
1/13/2024	Product C	Office Supplies	East	502
1/14/2024	Product A	Electronics	West	897
1/15/2024	Product E	Furniture	West	277
1/16/2024	Product A	Electronics	East	715
1/17/2024	Product E	Furniture	South	520
1/18/2024	Product A	Electronics	South	196
1/19/2024	Product C	Office Supplies	South	231
1/20/2024	Product A	Electronics	East	539

The data in the preview has been truncated due to size limits.

Extract Table Using Examples | Load | Transform Data | Cancel

Select the Text/CSV file given and load.

PART 1 | Step 3: Verify Data in Data View

Untitled - Power BI Desktop

Search

Sign in

File Home Help Table tools Column tools

Share

Name: Sales

Data type: Decimal number

Format: Whole number

Summarization: Sum

Data category: Uncategorized

Sort by column

Data groups

Manage relationships

New column

Structure

Formatting

Properties

Sort

Groups

Relationships

Calculations

Date	Product	Category	Region	Sales
Sunday, January 21, 2024	Product D	Electronics	East	800
Friday, February 2, 2024	Product D	Electronics	West	808
Sunday, February 11, 2024	Product D	Electronics	West	760
Thursday, February 15, 2024	Product D	Electronics	West	311
Sunday, March 3, 2024	Product D	Electronics	West	679
Saturday, March 16, 2024	Product D	Electronics	West	512
Sunday, March 17, 2024	Product D	Electronics	North	628
Tuesday, March 26, 2024	Product D	Electronics	West	647
Friday, March 29, 2024	Product D	Electronics	West	421
Saturday, April 13, 2024	Product D	Electronics	North	473
Friday, April 19, 2024	Product D	Electronics	East	706
Sunday, May 5, 2024	Product D	Electronics	North	280
Sunday, May 12, 2024	Product D	Electronics	South	835
Tuesday, May 14, 2024	Product D	Electronics	West	189
Saturday, May 18, 2024	Product D	Electronics	East	486
Saturday, May 25, 2024	Product D	Electronics	North	770
Sunday, May 26, 2024	Product D	Electronics	South	740
Tuesday, June 4, 2024	Product D	Electronics	West	830
Thursday, June 6, 2024	Product D	Electronics	North	711
Saturday, June 8, 2024	Product D	Electronics	East	743
Sunday, June 9, 2024	Product D	Electronics	West	602

Table: BI_LW1_Basic_Sales_Data (400 rows) Column: Sales (319 distinct values)

Data

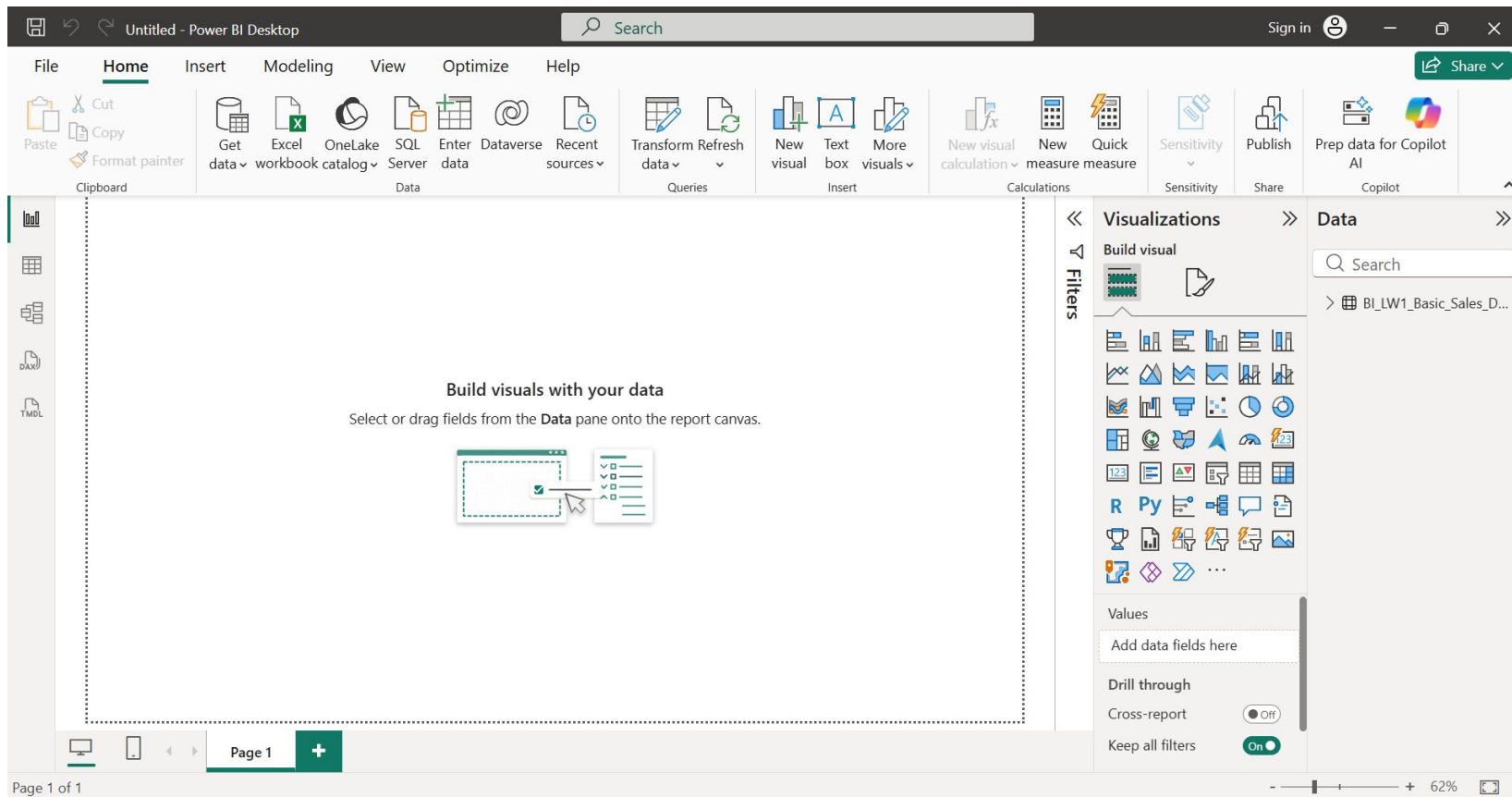
Search

BI_LW1_Basic_Sales_Data

- Category
- Date
- Product
- Region
- Σ Sales

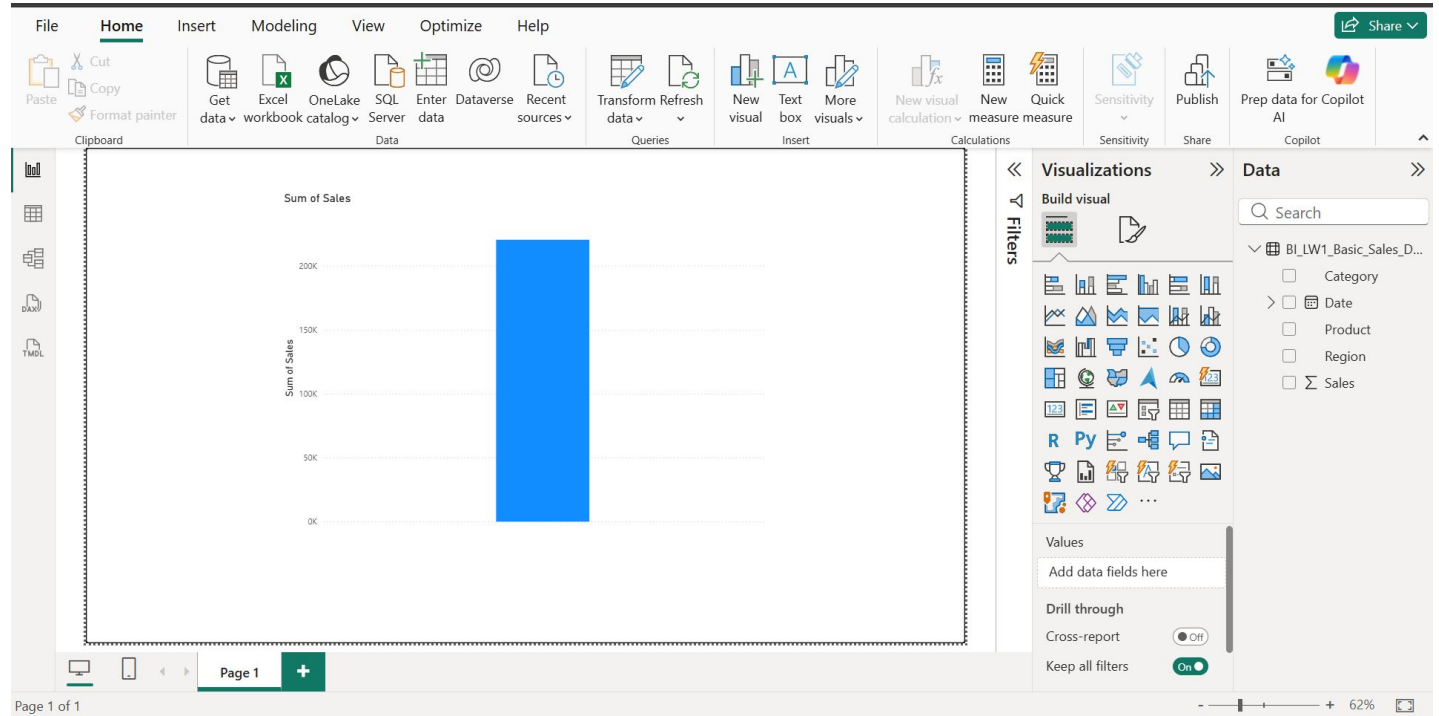
Check if data types are correct and columns visible.

PART 2: Exploring the Interface



PART 3 | Step 1: Quick Visualization

Power BI automatically creates a visual



Question:

- What type of chart was created?

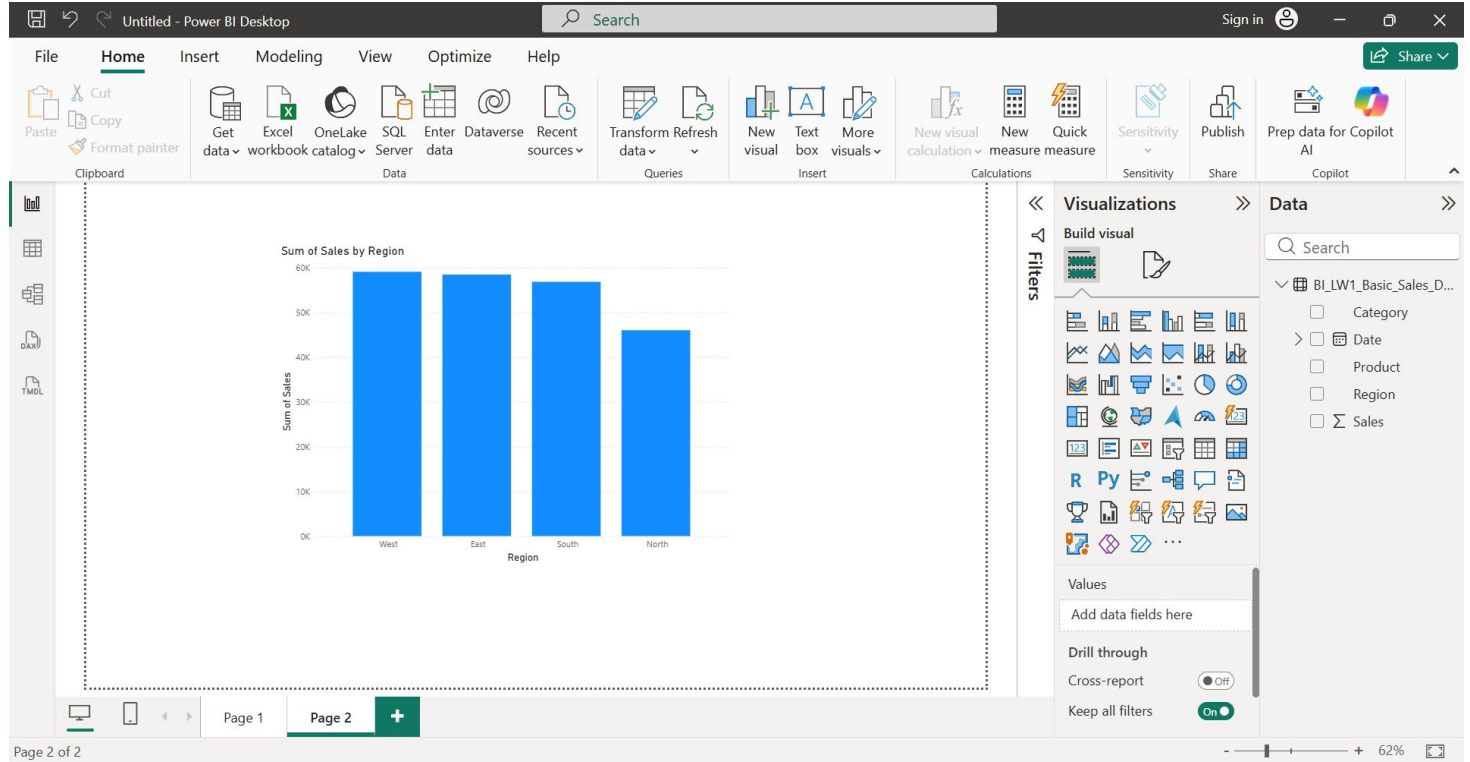
Answer: **Power BI shows a clustered column chart when you drag a numeric field like Sales onto the canvas.**

- What does it show?

Answer: **It shows the total (Sum) of Sales for the entire dataset.**

PART 3 | Step 2: Create a Sales by Region Chart

Clustered Column Chart of Sales per Region



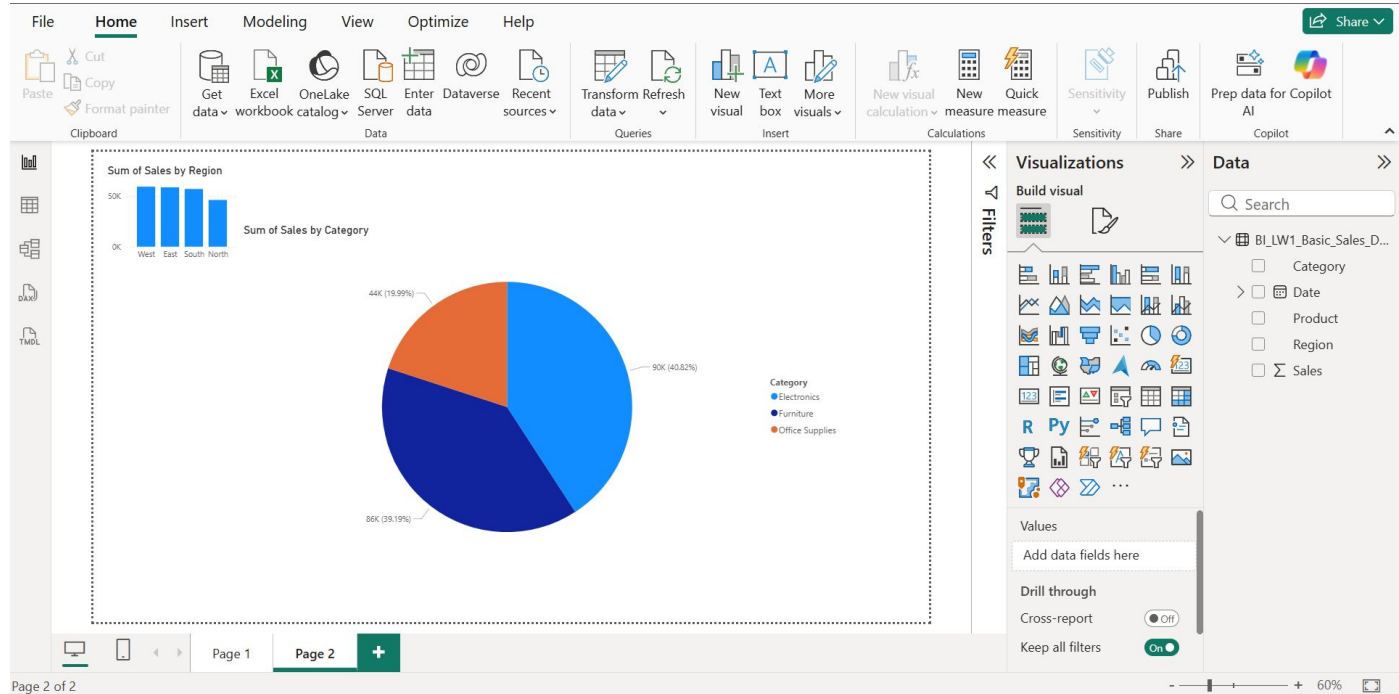
Question:

- Which region has highest sales?

Answer: **WEST** Region has the highest sales

PART 3 | Step 3: Sales by Category

Pie Chart of Category and its corresponding total sales



Question:

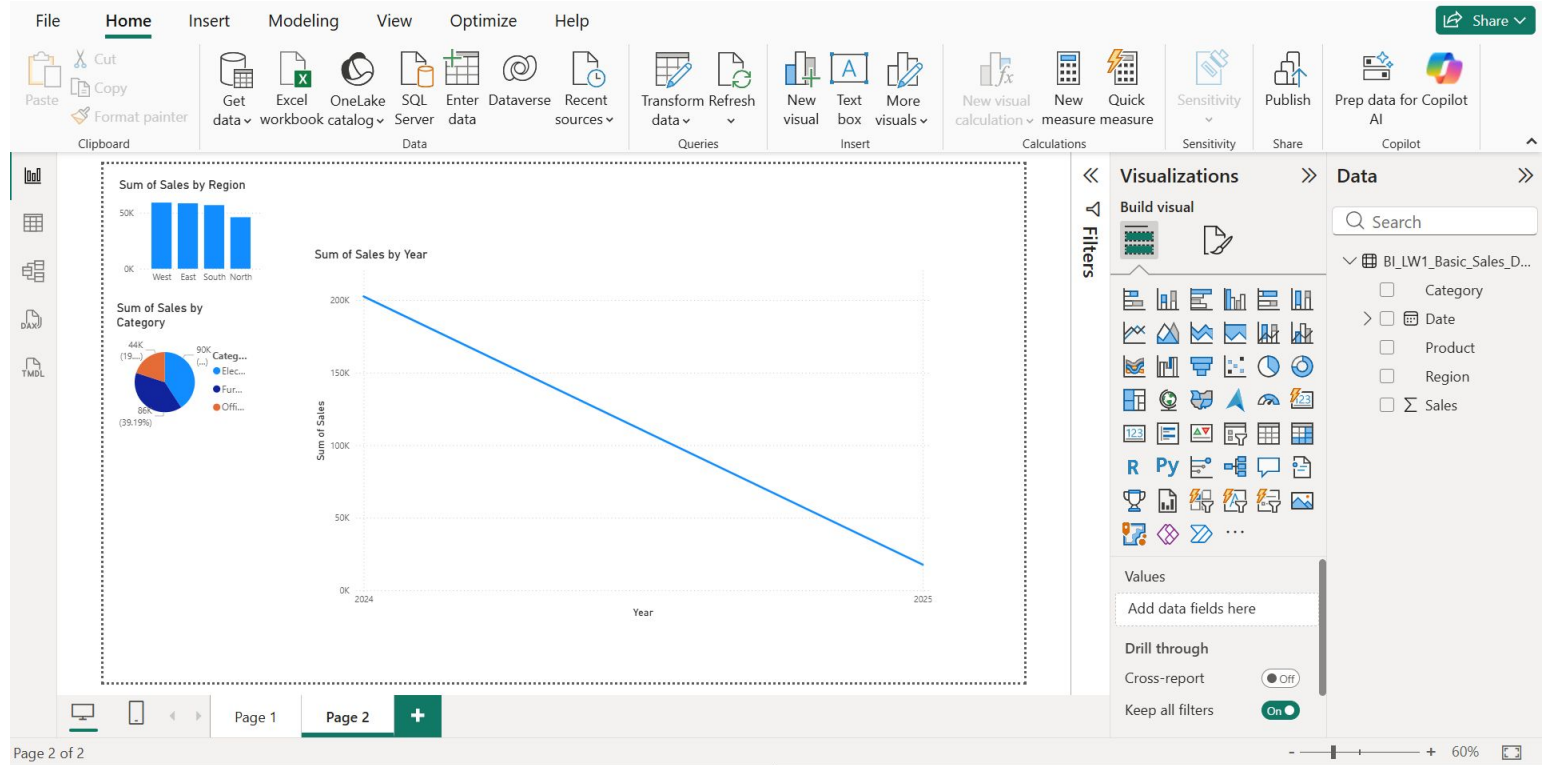
- Which category dominates?

Answer: **Electronics** dominate the sales with over 90K or 40.82% of the overall sales.

- Is the distribution balanced?

Answer: **Mostly Balance/ Not Balance** but not perfectly as two categories (Electronics and Furniture) are fairly balanced with each other, but Office Supplies lags behind, so overall the distribution is slightly skewed, **not fully balanced**.

PART 3 | Step 4: Sales Over Time



Line Chart of Sales within a period of time

Question:

- Is there growth? Answer: **No. There is no growth shown in the chart.**
- Any noticeable trend? Answer: **Yes — a downward (declining) trend.**

PART 4: Basic Data Insight Interpretation

Question:

- **Which region contributes most revenue?**

Answer: The *West region* is the top revenue contributor, though the differences between regions are not very large.

- **Which product category performs best?**

Answer: *Electronics* is the best performing category.

- **Are sales consistent across dates?**

Answer: *No* , sales are not consistent.

- **What business recommendation can you suggest?**

Answer: Maintain strong marketing and customer retention strategies in West Region and focus more in Electronics. Investigate also the sales drop in 2025 for future improvement. Boost weaker regions by regional promotions or localized strategies

LABORATORY QUESTIONS

Part A – Technical Questions

1. What are the five columns in the dataset?

Answer: **DATE, PRODUCT, CATEGORY, REGION and SALES.**

2. What data type is assigned to the “Sales” column?

Answer: **DECIMAL NUMBER**

3. Which Power BI view allows you to see raw data?

Answer: **DATA VIEW**

4. What chart type is best for showing trends over time?

Answer: **LINE CHART**

5. What aggregation is automatically applied to Sales?

Answer: **SUM OF SALES**

LABORATORY QUESTIONS

Part B – Analytical Questions

6. Which region has the highest total sales?

Answer: **WEST REGION** has the highest total sales.

7. Which category has the lowest performance?

Answer: **OFFICE SUPPLY** has the lowest performance.

8. Are sales increasing, decreasing, or stable?

Answer: **DECREASING**

9. If you were a manager, which region would you prioritize?

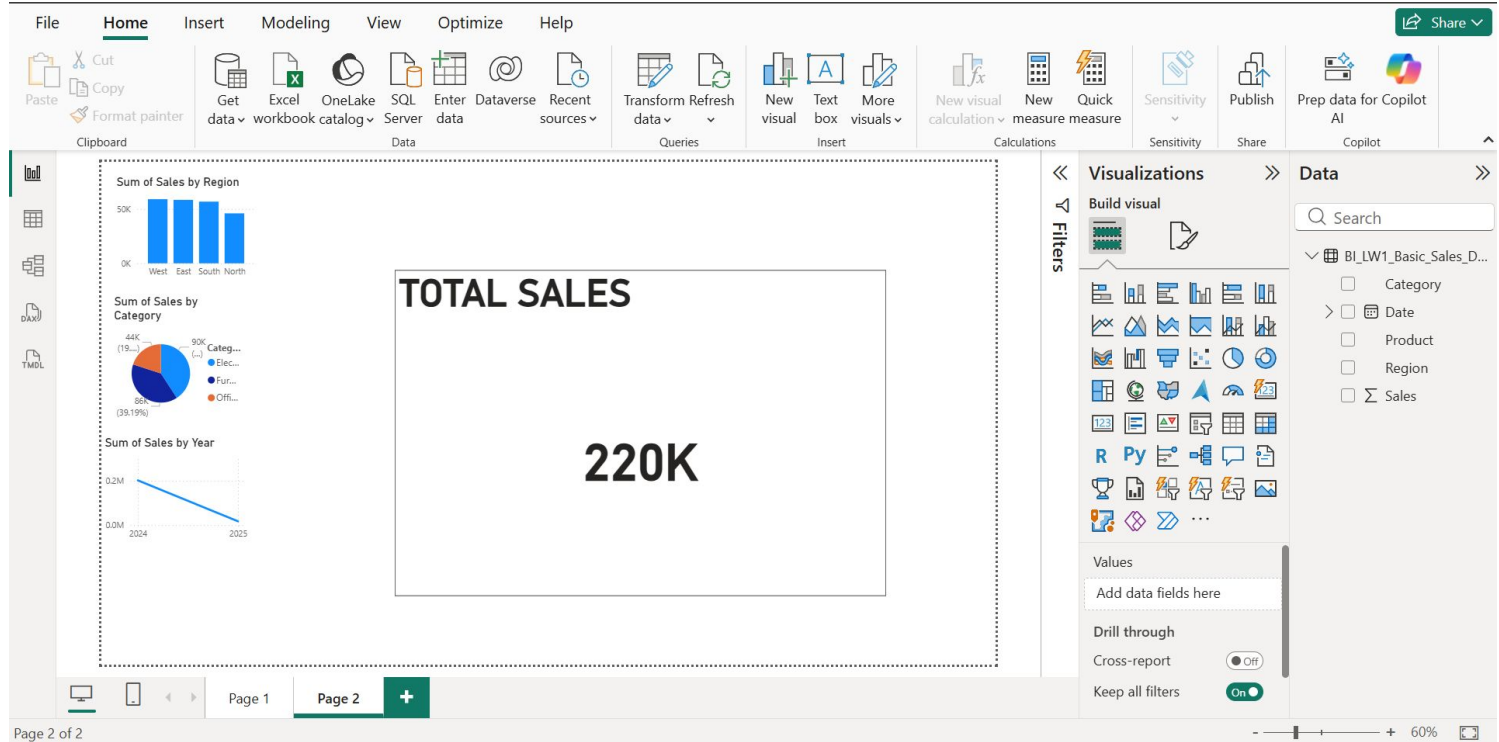
Answer: **WEST REGION**, as it has the highest revenue contributor and has strong marketing performance.

10. Provide one actionable recommendation based on the data.

Answer: **Focus marketing and inventory investment on Electronics in the West region.**

ENHANCEMENT SECTION | Task 1: Add a Card Visualization

Card Visualization
of Total Sales



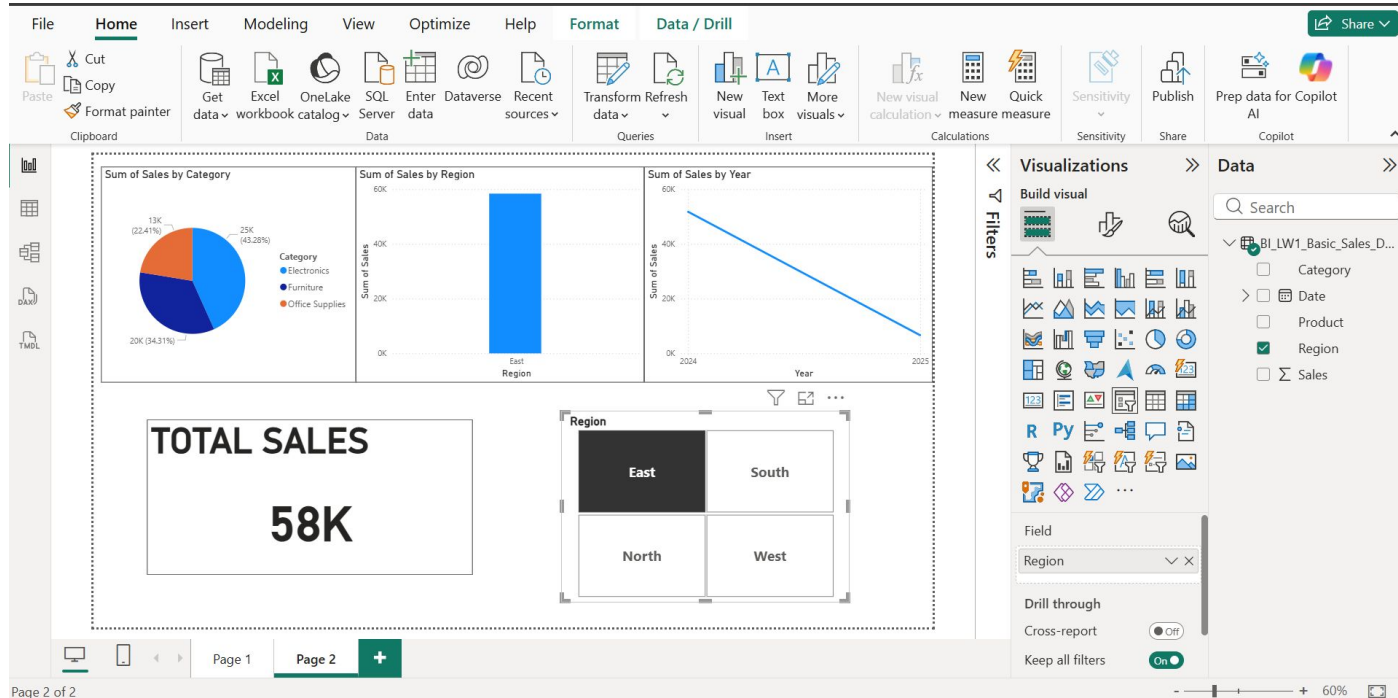
Question:

- What is the total sales amount?

Answer: **Total Sales is 220K**

ENHANCEMENT SECTION | Task 2: Add Slicer

Test Filtering of Sales per Region



Question:

- What happens to other visuals when you click a region?

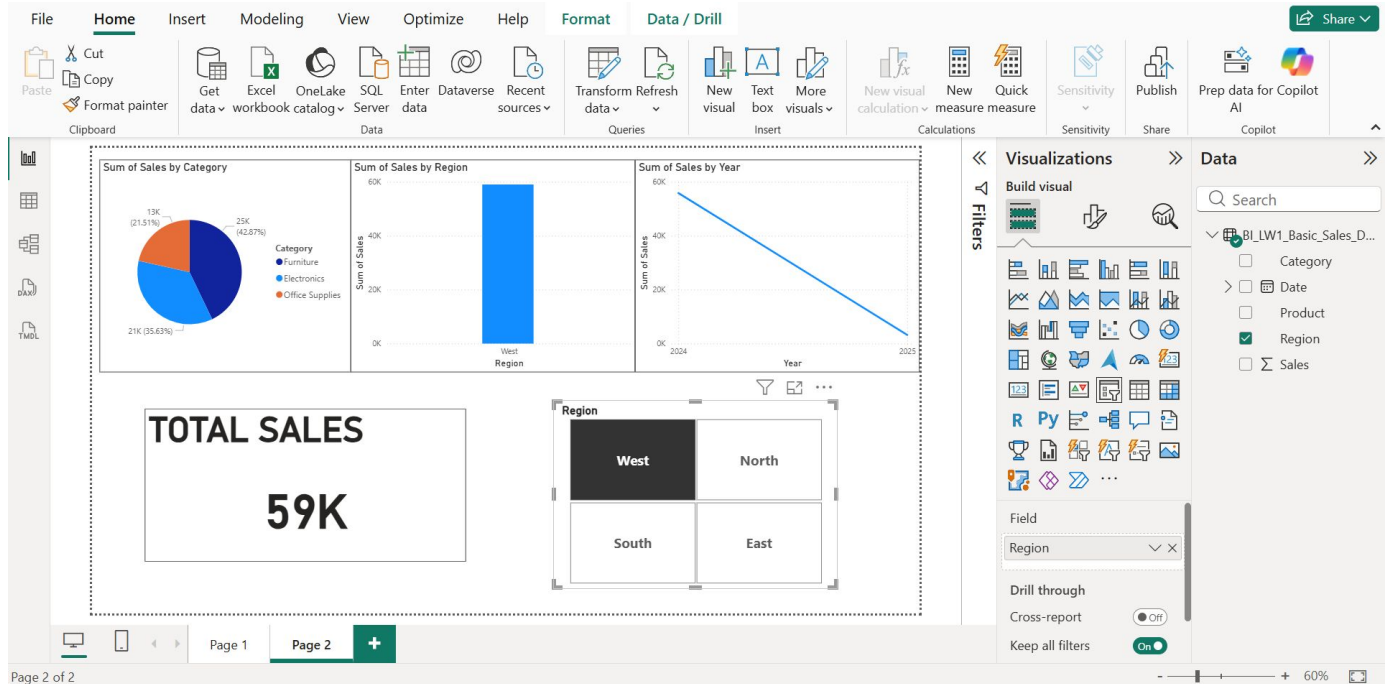
Answer: **All other visuals update automatically to show data only for the selected region.**

- Why is filtering important in BI?

Answer: **It is important in enable to focus analysis and helps identify regional, product or time-based paterns for users to compare segments easily.**

ENHANCEMENT SECTION | Task 3: Sort Sales

Sorting Sales by
Descending Options



Question:

- Does sorting improve readability?

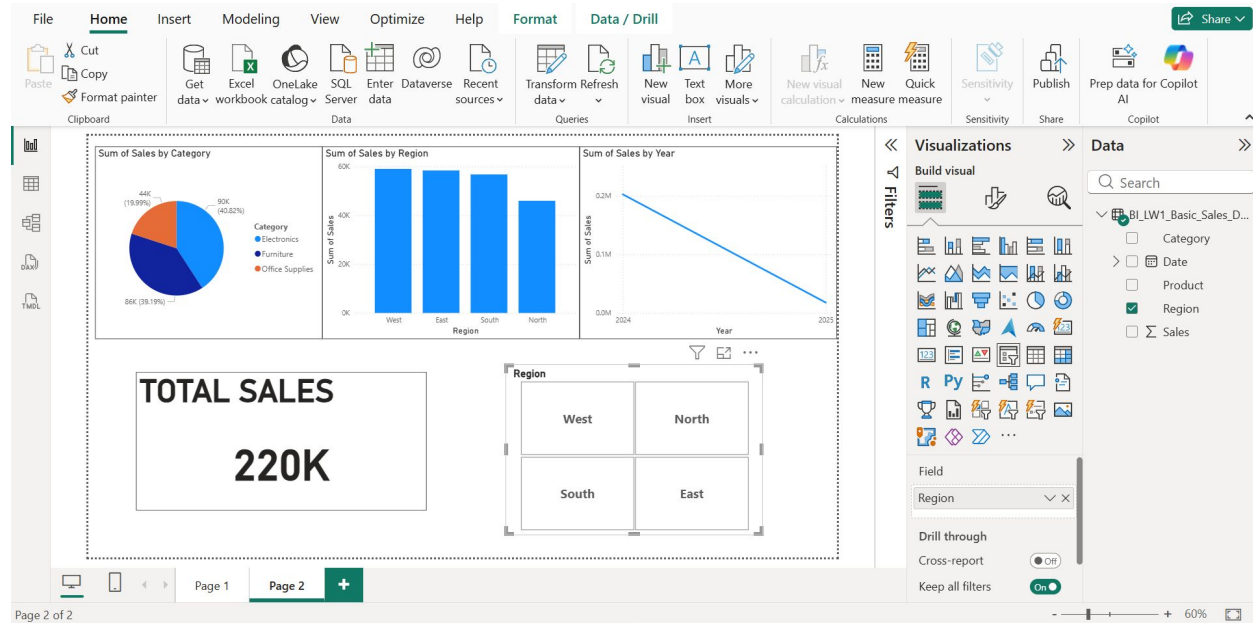
Answer: **YES. It improves sorting readability.**

- Why?

Answer: **For better and easier comparison and reduces time in interpreting performances.**

ENHANCEMENT SECTION | Task 4: Identify Outliers

Highest and Lowest Performance Region



- Which region is significantly higher or lower?

Answer: **WEST has the highest and NORTH as the lowest performance.**

- What might explain that difference?

Answer: **It might be because of the higher customer demand or population in the West or its stronger marketing and sales channels that leads to better product availability and distribution while the North has it otherwise.**

INSIGHT SUMMARY

This laboratory activity demonstrated how Power BI can transform raw sales data into meaningful business insights through visualizations. Using a CSV dataset containing Date, Product, Category, Region, and Sales, various charts and interactive tools were created to analyze overall performance and trends.

The West region recorded the highest total sales, making it the strongest market, while the North region had the lowest sales, indicating possible weaknesses in demand, distribution, or marketing. In terms of product categories, Electronics dominated total sales with over 90K (around 40.82%), making it the main revenue driver, while Office Supplies had the lowest contribution and may require improvement strategies.

The sales trend analysis showed a major decline from 2024 to 2025, suggesting unstable performance that requires further investigation. To improve results, the company should strengthen strategies in the West region and Electronics category while addressing weak performance in the North region and Office Supplies. Lastly, the slicer feature highlighted Power BI's dynamic filtering capability, allowing faster and more focused analysis across different regions and categories.